PROJECT SCOPE DOCUMENT

MBA 666

Green Hydrogen Production Plant

Submitted by:
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Project Title: Green Hydrogen Production Plant — 1 MW (Pilot)
Project Sponsor / Owner: Shrimi Agrawal
Project Manager & Registration ID: Shrimi Agrawal , MBA666_221036
Project Type: Individual academic project
Date: 03-11-2025

1. Project Overview

This project aims to design, procure, and install a small-scale **$1 MW Alkaline Electrolysis-based Green Hydrogen Production Plant**. The plant will produce $0.5 tons (480 kg) of green hydrogen per day using a hybrid power supply that prioritizes solar photovoltaic (PV) energy and utilizes a Battery Energy Storage System (BESS) for high renewable power utilization. The project directly supports India's **National Green Hydrogen Mission** (2023)

2. Objectives

The project will be deemed successful upon achieving the following measurable objectives:

- **Production Capacity:** Commission the plant to consistently produce **≥ 450 kg H2 per day** (equivalent to approx 20 kg/hour at 100 % capacity factor).
- **Renewable Energy Utilization:** Achieve an average **≥ 80 % renewable power utilization** across the 24/7 operational cycle through PV and BESS integration.(This is achieved by combining on-site solar power with Green Power Purchase Agreement (PPA) for the balance of grid power).
- **Financial Constraint:** Commission the facility within **12 months** at an estimated total CAPEX **below ₹ 25 crore.**

- **Safety & Compliance:** Ensure full design, installation, and operational compliance with Petroleum and Explosives Safety Organisation guidelines for high-pressure hydrogen storage.

3. Project Deliverables

Deliverables	Description
1 MW Electrolyzer Unit	Installed Alkaline Electrolyzer system (including rectifier and balance-of-plant) capable of H2 purity **≥ 99.999 %**.
Hybrid Power System	Integrated **2.5 MWp** Solar PV array and **1 MWh BESS** for buffered power supply.
Hydrogen Storage	High-pressure **(350 bar)** cascade cylinder storage system with minimum **3 days** capacity **(approx 1.5 tons).**

4. Scope Definition

Inclusions:

- Procurement, installation, and testing of electrolyzer system
- Integration with 2.5 MWp solar plant,including land preparation and mounting structure.
- Integration of the 1 MWh **Battery Energy Storage System (BESS)** for power smoothing and demand response.
- Construction of utility buildings and storage facilities
- 3-year operations and training support

Exclusions:

- Hydrogen transport pipelines
- Large-scale ammonia conversion units
- Long-term O&M beyond 3 years

5. Key Base Conditions

Parameter	Value	Source
Preferred Location	Gujarat (Coastal	High solar irradiance (1

	Industrial Zone)	825 kWh/kWp/yr), port access, low land cost
Project Lifespan (LCOH)	20 Years	Standard industrial project horizon for LCOH calculation.
Discount Rate	10 %	Representative value for Weighted Average Cost of Capital for an RE project in India.
Electricity Cost (Grid)	₹ 4 / kWh	Gujarat renewable tariff (MNRE data) for grid backup/supplement.
Water Cost	₹ 40 / m^3	GIDC industrial tariff (Gujarat Water Supply 2025) approx ₹ 0.04 /L
H_2 Energy Consumption	50 kWh / kg H2	Benchmark for Alkaline Electrolyzer (IRENA 2024)
Annual Operating Days	365 days at 100% capacity factor(CF)	Aim for 24/7 continuous operation,facilitated by BESS and grid power

6. Detailed Calculations and Preliminary Cost Estimate

6.1 Technical Calculations (Volume & $ RE Matching)

Calculation	**Formula / Rationale**	**Value**
Daily H_2 Production	1 MW × (1000 kW/MW) / (50 kWh/kg) × 24 h/day	480 kg/day (≈ 0.48 tons/day)
Annual H_2 Production	480 kg/day × 365 days/year	175.2 tons/year
Daily Power Requirement	1 MW × 24 hours	24 MWh/day

Solar PV Daily Output	2.5 MWp × 24 h/day × 20% CF	12 MWh/day
Grid Energy Supplement	24 MWh (Req) – 12 MWh (Solar)	12 MWh/day
Daily Water Requirement	9 L H_2O/kg H_2 × 480 kg/day	4,320 Litres/day (≈ 4.3 m^3/day)
RE-Backed Grid Supplement	24 MWh(Req) - 12 MWh(Solar)	12 MWh/day

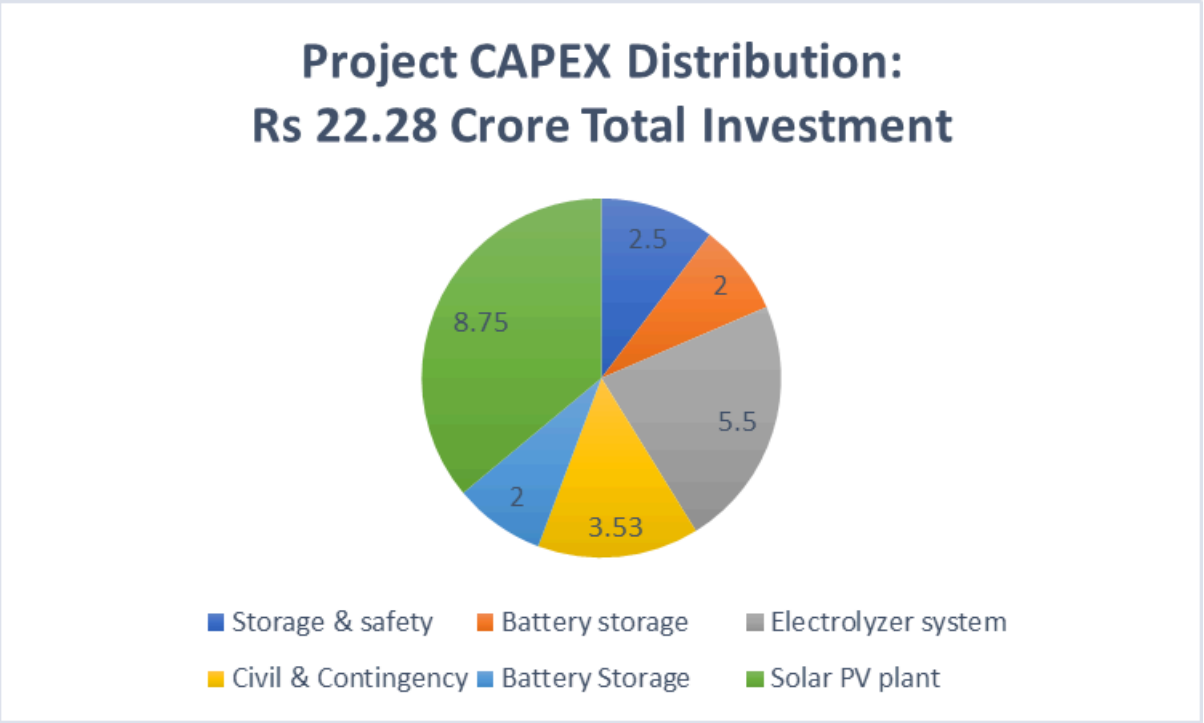
6.2 Preliminary CAPEX Estimate

The estimate includes the BESS cost based on the search result, which indicates a cost of approx ₹ 1.7 million (INR 17 lakhs) per MWh in 2024 (scaled up to ₹ 2.0 crore for a complete 1 MWh system inclusive of BoS and soft costs for 2025).

Component	Unit Cost	Capacity	Total Cost (₹ crore)	Calculation / Rationale
Electrolyzer System	₹ 55,000 / kW	1,000 kW	5.50	1000 kW x 55,000/10^7
Solar PV Plant	₹ 35,000 / kW	2,500 kW	8.75	2500 kW x 35,000/10^7
Battery Energy Storage System (BESS)	₹ 2.0 crore/ MWh	1 MWh	2.00	Based on 2025 merchant BESS CAPEX benchmark.
Storage & Safety Systems	—	—	2.50	To account for high-pressure 350 bar storage and PESO compliance.
Civil & Utility Works	—	—	1.50	Water treatment,compressors,buildings,etc.

EPC, Contingency, Training	10% of subtotal	—	2.03	10 % of (₹ 18.75 crore)
Total Estimated CAPEX			22.28 crore	Within the ₹25 crore budget

The estimated costs align with 2025 market projections for India's National Green Hydrogen Mission (NGHM) targets. The Electrolyzer cost (INR 55,000/kW) is commercially sound for **Alkaline** technology in the 1 MWclass. The **BESS** component, priced at INR 2.0 crore}/MWh, reflects the mid-range cost for a 4-hour installed system, which is crucial for achieving high utilization and stabilizing the variable solar input. The Solar PV cost is standard for utility-scale deployment, benefiting from economies of scale.



6. Detailed Calculations and Preliminary Cost Estimate

1.Electrolyzer Cost: How can India Indigenise Hydrogen Electrolyser Manufacturing? CEEW
2. Global Cost Benchmarks and targets: How can India Indigenise Hydrogen Electrolyser Manufacturing? CEEW
3.CAPEX for Solar PV Plant:25kW Solar Panel System Price in India(2025), Subsidy & More
4. Grid BESS cost:Battery Energy Storage Systems (BESS): The Future of Energy Storage in India (2025 Perspective)